



Review

The clinical utility of anti-double-stranded DNA antibodies and the challenges of their determination

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ABSTRACT

Autoantibodies against double-stranded DNA (dsDNA) were first described > 60 years ago and although they are still one of the most clinically relevant autoantibodies, test results may be more challenging to interpret compared to other autoantibody tests. They are a serological hallmark of systemic lupus erythematosus (SLE) and are included in both the American College of Rheumatology (ACR) and the Systemic Lupus International Collaborating Clinics (SLICC) classification criteria for SLE. Furthermore, anti-dsDNA antibodies (a-dsDNA) have been shown to associate with SLE disease activity and coincide with renal involvement. Given their importance and long history, one might assume that immunological tests for a-dsDNA are standardized and give comparable results. However, even though there has been an international reference standard serum (the WHO Wo/80), different methods for the detection of a-dsDNA and tests from different manufacturers give different results for the same samples. This disparity is due to the diversity of possible antibodies generated to this biochemically complex antigen, which may have different clinical associations. The goal of this review is to summarize the current knowledge regarding the clinical associations with a-dsDNA, highlight challenges in a-dsDNA testing, and elucidate the reasons for discrepant results between methods or manufacturers.

1. History of anti-dsDNA antibodies

The history of anti-dsDNA antibodies (a-dsDNA) is closely linked to the history of systemic lupus erythematosus (SLE) research. In 1948 Malcolm Hargraves and colleagues were the first to describe the LE cell (Hargraves, 1969), a bone marrow-derived cell with a characteristic morphology that was found circulating in SLE patients. Nine years later, in 1957, further research on LE cells led to the discovery by Holman and Kunkel that deoxyribonuclease destroyed the antigenic determinant in nucleoprotein that was a requisite for the LE cell phenomenon (Holman and Kunkel, 1957). Extending this observation, they suggested that antibodies to DNA in the sera of SLE patients were involved in this process. At about the same time, several researchers independently confirmed the presence of these antibodies in SLE patients (Robbins et al., 1957; Miescher and Strassle, 1957; Seligmann, 1957). These findings triggered intense research on a-dsDNA, their nature and clinical relevance. In 1966, Tan and colleagues hypothesized that there may be a link between the presence of a-DNA and the pathogenesis of renal lesions in cases with SLE (Tan et al., 1966). One year later this was impressively confirmed by Koffler and colleagues, who showed that a-

dsDNA could be detected in the eluates of renal biopsies from patients suffering from lupus nephritis (LN) (Koffler et al., 1967). At that time, a-dsDNA were seen as a distinct entity, but no discrimination was made between e.g. antibodies to double- or single-stranded DNA. Finally, in 1982 a-dsDNA were included in the revised ACR criteria for the classification of SLE but without specifying immunoglobulin isotype/isotypes or method used to detect them (Tan et al., 1982). These criteria were updated in 1997 without any changes regarding a-dsDNA (Hochberg, 1997), and 15 years later a-dsDNA were also included in the classification criteria for SLE of the Systemic Lupus International Collaborating Clinics (SLICC) (Petri et al., 2012). The SLICC classification differentiates between a-dsDNA levels measured by ELISA from those measured by other methods. For ELISA, levels twice the laboratory cut-off should be considered as positive (Fig. 1).

2. Different anti-DNA antibody populations have different clinical associations

The expression a-dsDNA comprises a mixture of antibodies that are all directed to DNA (Fig. 2), but differ in their associations with disease

Abbreviations: ACR, American College of Rheumatology; a-dsDNA, anti-dsDNA; SLE, systemic lupus erythematosus; SLICC, Systemic Lupus International Collaborating Clinics

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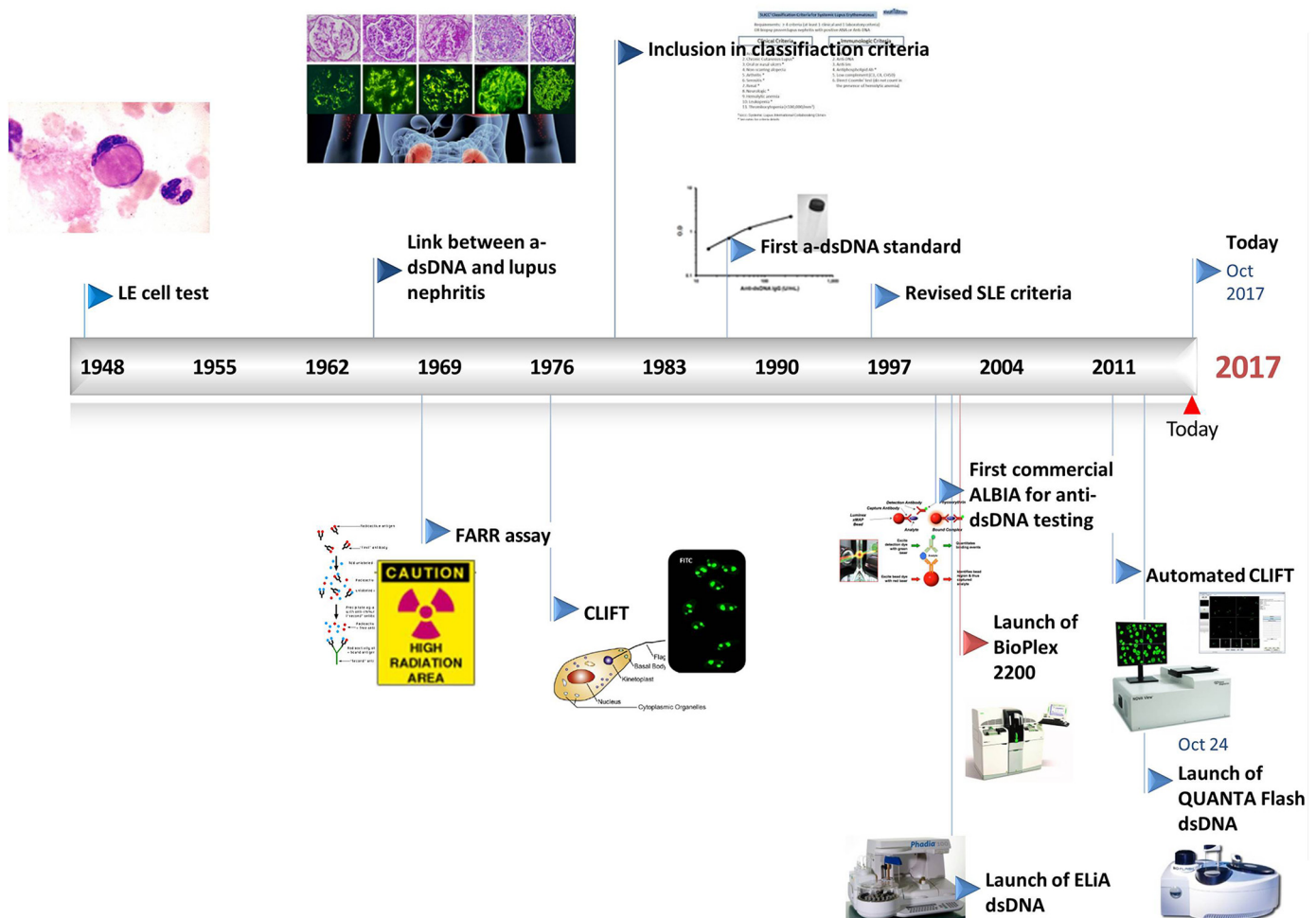


Fig. 1. History of anti-dsDNA antibodies (a-dsDNA) from discovery to current use. The history of a-dsDNA started with the discovery of the LE cell. Later on, several scientific findings underlined the importance of a-dsDNA in systemic lupus erythematosus (SLE) and triggered the development of several test methods which evolved in terms of clinical performance as well as automation.

or clinical symptoms (Pisetsky, 2016). Antibodies specifically directed against DNA bases (i.e. guanosine, thymidine, adenosine, cytosine) typically detect epitopes on single-stranded DNA (ssDNA). These antibodies have been described in a wide range of systemic autoimmune rheumatic diseases (SARD), such as SLE, rheumatoid arthritis, mixed connective tissue disease, undifferentiated connective tissue disease, and systemic sclerosis (Koffler et al., 1973; Locker et al., 1977), as well as in unrelated diseases, e.g. in primary biliary cholangitis, chronic hepatitis and ulcerative colitis and infectious diseases (Locker et al., 1977; Tsuchiya et al., 1994; Pisetsky, 2016). The second population of a-dsDNA is directed against the sugar-phosphate backbone of DNA, which is present in both ssDNA and dsDNA (Arana and Seligmann, 1967). Thus, anti-ssDNA and a-dsDNA can overlap based on binding to this shared DNA epitope. Since these antibodies are detected by an a-dsDNA assay, they are primarily believed to be specific for SLE, although there are some limitations that will be explained below. The third a-dsDNA population is directed to the dsDNA double helix, and the epitopes recognized by them appear to be conformational with no apparent cross-reactivity with ssDNA antibodies (Arana and Seligmann, 1967). As with the second population of a-dsDNA, these antibodies are believed to be specific for SLE (Koffler et al., 1969a; Koffler et al., 1969b). Since dsDNA can occur in different conformations (β -helical dsDNA, Z-DNA/left handed DNA, DNA/RNA hybrids, 'kinked' dsDNA, triplex DNA), the antibodies binding the double helix can be further divided into other subpopulations. Early studies, particularly those of Stollar and his colleagues, focused on the heterogeneity of the DNA structures and

epitopes that comprised a rather wide spectrum of targets including native dsDNA, left-handed or Z-DNA, triple helical DNA, and DNA/RNA hybrids (Rekvig, 2015; Stollar and Raso, 1974; Lafer et al., 1983; Lafer et al., 1981; Busen et al., 1982). It is still unclear, if these subpopulations have different clinical associations.

When looking at the whole entity of a-dsDNA, the most striking difference between antibody subpopulations, which is also connected to different clinical phenotypes, is antibody affinity or avidity, respectively. It's mostly the high avidity antibodies that have been suggested to be specific for SLE, to follow disease activity, and to be associated with LN (Werle et al., 1992; Andrejevic et al., 2013). In contrast, the low avidity a-dsDNA are not specific for SLE, but occur in a variety of other diseases, among them connective tissue disease, as well as unrelated diseases (Infantino et al., 2015; Smeenk et al., 1988). Furthermore, they do not seem to follow disease activity, and are not associated with renal involvement. Thus, their clinical utility for SLE diagnosis, follow-up and risk of nephritis is limited. However, most data on the affinity/avidity of anti-dsDNA antibodies are derived from high salt conditions during incubation with serum samples, which might not only suppress low avidity interactions, but might also alter the properties of the DNA. Unfortunately very little affinity data are available on alternative approaches such as competitive ELISAs or surface plasmon resonance (SPR) (Buhl et al., 2007; Buhl et al., 2009; Fiegel et al., 2010). The weak association of low avidity a-dsDNA might be a reason for tests such as the enzyme-linked immunosorbent assay (ELISA) showing a lower specificity for SLE than e.g. the *Crithidia luciliae*

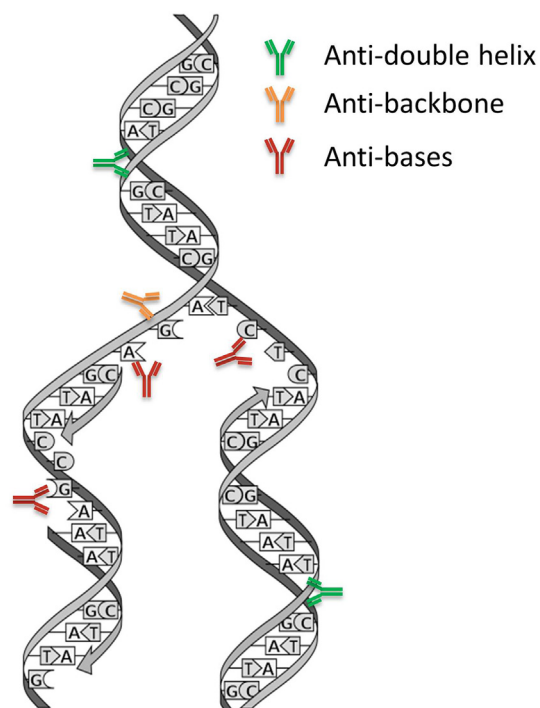


Fig. 2. Different populations of antibodies to DNA. Pure anti-ssDNA antibodies (red), anti-ssDNA and anti-dsDNA antibodies (orange), and pure anti-dsDNA antibodies (green). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

immunofluorescence test (CLIFT) (Villalta et al., 2002; Villalta et al., 2003) or the chemiluminescence immunoassay (CLIA).

3. Different methods for anti-dsDNA antibody detection

Assays for the detection of a-dsDNA have significantly evolved from the time when they were first described in 1948 (Pisetsky, 2016). Today a variety of different a-dsDNA test technologies are available (Pisetsky, 2013; Mahler and Fritzler, 2007), as shown in Table 1.

Some of the early techniques, such as the Ouchterlony immunodiffusion test (Ouchterlony, 1949) or complement fixation (Barnett, 1968) are still in use in some specialty laboratories, but have been widely replaced by more advanced, more sensitive, quantitative and less labor intensive methods. Radioactive assays, such as the Farr assay (Pincus et al., 1969), the PEG precipitation assay (Riley et al., 1979), or the filter binding assay (Kredich et al., 1973; Ginsberg and Keiser, 1973) have mostly been abandoned by laboratories, mainly due to the constraints linked to availability of the radiolabelled dsDNA, as well as issues relating to safety and handling radioactivity.

Nevertheless, the Farr assay, first described for the detection of a-dsDNA in 1969, is sometimes still regarded as the gold standard because of its high specificity for SLE. In addition, the Farr assay is the method that is included in the widely used composite index SLE disease activity index 2000 (SLEDAI-2K) (Gladman et al., 2002). The specificity for SLE is attributed to the use of high salt (ammonium sulphate) concentrations during the precipitation step, which is believed to select for high avidity antibodies (Mahler and Fritzler, 2007). However, apart from relying on a reliable source of radioactive dsDNA, there are other disadvantages. Due to the precipitation of antibodies without using a secondary antibody, it detects a-dsDNA of all immunoglobulin subclasses, i.e. IgG as well as IgM and IgA (Egner, 2000). Although all immunoglobulin isotypes can be found in various conditions other than SLE (e.g. infectious diseases, TNF inhibitor treatment) (Vaz et al., 2013; Vaz et al., 2016), IgM is more common in those conditions. (Charles et al., 2000). Consequently, Farr assay results can be misleading in

some cases, as the assay cannot differentiate between IgG and IgM. Whether or not IgM a-dsDNA provides value in identifying early SLE cases is an important question and deserves further research. Another aspect is that classification criteria do not specify which of the a-dsDNA Ig isotypes should be detected (Hochberg, 1997; Petri et al., 2012).

The development of non-radioactive methods resulted in a number of assays which all have their pros and cons. The indirect immunofluorescence test using the hemoflagellate *Crithidia luciliae* as a substrate (*Crithidia Luciliae* Immunofluorescence Test – CLIFT) was developed in 1975 by Lars Aarden and colleagues (Aarden et al., 1975; Crowe and Kushner, 1977) and is still widely used for the detection of a-dsDNA. The *Crithidia* cell contains a modified mitochondrion, the kinetoplast, which contains highly compacted native dsDNA. Although it is thought that the kinetoplast is composed of highly compacted dsDNA in the absence of histones and hence is an ‘ideal’ a-dsDNA target, some evidence points to a highly complex representation of dsDNA attended by histone-like proteins and other proteins involved in regulating the expression of kinetoplast DNA (Lukes and Maslov, 2000; Avliyakov et al., 2004). Nevertheless, with some exceptions (Compagno et al., 2013), the CLIFT is renowned for high specificity and positive predictive value for SLE and is the reason why it is often used for confirmation of a positive result obtained with another method. However, the diagnostic sensitivity for SLE of the CLIFT assay for the diagnosis of SLE is low (typically 20–35% depending on assay conditions, serum dilution and not all *Crithidia* kits perform identically), making it not suitable for a-dsDNA and SLE case finding (Slater et al., 1976; Mahler and Fritzler, 2007; Infantino et al., 2015). In addition, because the results are semi-quantitative, it is not a test that is easily amenable to following the clinical course and predicting flares of SLE. The recent introduction of automated microscopes for reading *Crithidia luciliae* slides has resulted in a more standardized assay approach, decreased subjectivity of interpreting CLIFT results, and is less time consuming (Lakos et al., 2016).

The apparent low sensitivity of the CLIFT assay, which can mostly be observed in early SLE (Enocsson et al., 2015), was initially addressed by ELISA technology (Kawai et al., 1982). These assays, even though their performance differs between manufacturers, usually have a high sensitivity (> 60%) for two reasons: first, even nanograms of antibodies can be detected by this method (Pisetsky, 2013; Pisetsky, 2016; Mahler and Fritzler, 2007), and second, they detect a spectrum of a-dsDNA antibodies with avidities ranging from low to high (Villalta et al., 2002). This makes ELISAs suitable for screening for a-dsDNA, but their low specificity for SLE due to the detection of low avidity antibodies results in the need of a confirmatory assay that is highly specific for SLE. In addition, historically, dsDNA was immobilized in many ELISAs using poly-L-Lysine (Fish and Ziff, 1981). However, this technical approach was proven to produce false positive results (Kroubousos et al., 1992). Some manufacturers have attempted to overcome this shortcoming of the ELISA technique by using a higher salt concentration in the wash buffer, resulting in so-called “confirmatory high avidity a-dsDNA ELISA” tests. However, in this context it remains unclear if the high salt concentration has an impact on the quality (antigenicity) of the dsDNA (Tan and Chen, 2006).

Since automation became increasingly utilized in laboratories, fully automated systems for the detection of autoantibodies were developed. The fluoroenzyme immunoassay (FEIA) is one (Hernando et al., 2002) that offers an a-dsDNA test that shows similar performance to ELISA assays, along with some probable improvement of specificity. Like ELISA, the FEIA uses dsDNA-coated polystyrene wells but the detection of bound antibodies is performed by fluorescence instead of an enzyme-generated colour reaction, thereby yielding a broader analytical measuring range (AMR), which in turn results in fewer dilutions of samples when results are above the AMR of the assay.

The addressable laser bead immunoassays (ALBIA) show a performance similar to ELISA assays (Shovman et al., 2005) and in one study also had disease specificity comparable to CLIFT (Enocsson et al.,

Table 1
Characteristics of anti-dsDNA antibody immunoassays.

Assay	Characteristics/Mode of Operation	Results
Double Immunodiffusion (Ouchterlony)	Precipitation of immune complexes in a gel. Detection of antibodies by visual inspection of precipitation bands. Low sensitivity.	Semi-quantitative
Counter-electrophoresis (CIE)	Immunoglobulins migrate to the negative end of a capillary system in an electric field and precipitation of dsDNA (modification of Ouchterlony).	Semi-quantitative
Complement fixation assay	Fixation of complement by immune complexes. Detection of antibodies by visual inspection of colour development.	Qualitative
Farr radioimmunoassay (RIA)	Precipitation of immune complexes using high ammonium sulfate concentrations. Detection of antibodies by measuring the radioactivity in the precipitated pellet. Typically used hard-emitting gamma-radiolabeled (i.e. P^{32}) dsDNA.	Quantitative
PEG (polyethylene glycol) precipitation assay	Precipitation of immune complexes using polyethylene glycol. Detection of antibodies by measuring the radioactivity of the pellet.	Quantitative
Filter binding assay	Capturing immune complexes by filtration on nitrocellulose filter. Detection of antibodies by measuring the radioactivity of the filter. Typically used soft-emitting beta-radiolabeled (H^3 /tritium, C^{14}) dsDNA	Quantitative
<i>Crithidia luciliae</i> immunofluorescence test (CLIFT)	Antibody binding to the kinetoplast of the hemoflagellate adhered to a glass slide. Detection of antibodies by indirect immunofluorescence using a fluorescent-dye coupled secondary anti-human IgG (or IgM, IgA) antibody.	Semi-quantitative
ELISA	Antibody binding to dsDNA antigen, coated onto 96-well polystyrene plates. Detection of antibodies by using an enzyme-coupled secondary anti-human IgG (or IgM, IgA) antibody and colour detection. Assay preparation typically required pre-coating the well with cationic poly-lysine to facilitate binding of anionic DNA	Quantitative
Fluoroenzyme immunoassay (FEIA)	Antibody binding to dsDNA coated onto polystyrene wells. Detection of antibodies by using an enzyme-coupled secondary anti-human IgG (or IgM, IgA) antibody and fluorescence detection.	Quantitative
Chemiluminescence immunoassay (CLIA)	Antibody binding to dsDNA antigen, covalently bound to paramagnetic beads. Detection of antibodies by using an isoluminol-conjugated secondary anti-human IgG (or IgM, IgA) antibody and chemiluminescence detection.	Quantitative
Addressable laser bead immunoassay (ALBIA)	Antibody binding to dsDNA antigen, covalently bound to laser-detectable coloured beads. Detection of antibodies by using a rhodamine-dye coupled secondary anti-human IgG (or IgM, IgA) antibody and laser/fluorescence detection.	Quantitative

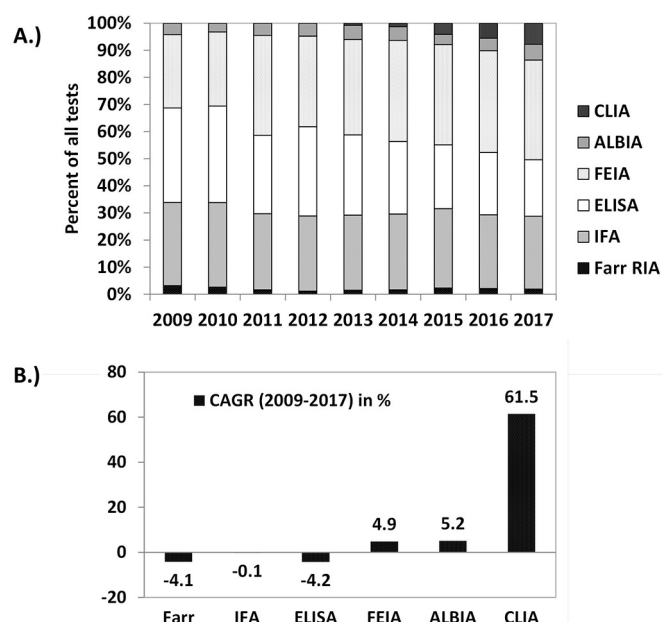


Fig. 3. Change in usage of methods for a-dsDNA testing for the years 2009–2017, based on the results of the UKNEQAS results for a-dsDNA sample distributions. In A.) The relative proportion of the different methods is shown (from 2009). In B.) the compound average growth rates (CAGR, in percent) are displayed.

2015). Their improvement is mainly in terms of workflow, high throughput and the flexibility of multiplexing with concurrent tests for other autoantibodies.

A more recently launched automated a-dsDNA test using the chemiluminescence immunoassay (CLIA) technique showed performance improvements compared to ELISA, FEIA and ALBIA by combining good sensitivity with high specificity for SLE (Infantino et al., 2015; Bentow et al., 2016). In a comparative study, the assay had the best sensitivity

(at a fixed desired specificity of 94%) when compared to two ELISAs, CLIFT and ALBIA (Infantino et al., 2015). In this assay, dsDNA is coupled to paramagnetic beads, a feature that is believed to remove low avidity antibodies by using a very stringent wash procedure, which increases the specificity of the assay while maintaining high sensitivity for SLE. Toh et al. reported that the CLIA correlates more closely with the Farr assay than other methods (Toh et al., 2014), and exhibits a good combination of sensitivity and specificity (~80%), providing excellent clinical value of the results. Even though the CLIA might be used as the only a-dsDNA assay in a routine setting, it may be advisable to verify questionable results (positive or negative) by a second method (i.e. a highly specific CLIFT), depending on the pre-test probability.

The development of new, non-radioactive and/or automated methods for a-dsDNA detection resulted in a shift in the usage of the different methods (Damoiseaux et al., 2014). However, this is very much dependent on the geographic location of the laboratory. A survey from 2014 showed that while the Farr assay almost disappeared from many European laboratories, and was used by only 15% of the laboratories in Israel, ELISA and FEIA are among the most widely used tests, ranging from 51% in France to 80% in Ukraine for ELISA, and from 56% to 80% in Finland for FEIA (Damoiseaux et al., 2014). The biggest difference in test usage was found for CLIFT, ranging from 0.0% in Ukraine to 94% in Sweden. The high number for CLIFT in Sweden is attributed to the use of two different assays in most laboratories. The switch towards automated a-dsDNA tests, which has occurred over the past few years, can also be demonstrated by looking at the results of the UK External Quality Assessment Scheme (UKNEQAS), a proficiency testing organization with many international, non-UK users, as well. When evaluating the first sample distribution of each year for a-dsDNA testing for the years 2009–2017, it can be clearly seen that while Farr RIA and CLIFT are quite stable in the relative proportion of all participants, there is a clear decrease in ELISA usage in preference to more recent, automated methods, such as FEIA and especially CLIA (Fig. 3), with fastest growing usage of the CLIA tests.

4. Why different methods for anti-dsDNA antibody detection give different results

In order to standardize a-dsDNA assays, an international reference serum (the Wo/80), which was assigned an activity of 200 IU/ml, was made available in 1988 (Feltkamp et al., 1988; Feltkamp, 1990). This reference standard was widely adopted and many quantitative assays such as Farr RIA, ELISA, FEIA, ALBIA, have been calibrated against it. However, the expectations that the reference standard would help make different assays produced by different manufacturers more comparable have not been realized. An explanation of why different a-dsDNA assays give comparable results of 200 IU/ml for the Wo/80 reference standard but give different results for other patient samples suggests an unmet challenge in standardizing this assay. The Wo/80 was derived from a single patient serum with a polyclonal B-cell response. All assays for the detection of a-dsDNA, even if calibrated against the Wo/80, differ in terms of methodology, antigen source, buffer composition, conjugate used (if any), and other assay conditions (Meroni et al., 2014; Sheldon and Dellavance, 2015), leading to different performance. Consequently, there is no a-dsDNA assay which is able to detect all of the antibody populations present in the Wo/80 standard, but each assay will only detect certain a-dsDNA subpopulations. However, even if only subpopulations of the Wo/80 are detected, all these different assays give the same result by definition: 200 IU/ml, because they are calibrated against the standard. This implies that the different assays may under- or overestimate certain subpopulations of the a-dsDNA contained in the Wo/80 standard, in order to obtain the desired result of 200 IU/ml. As a consequence, a serum from another SLE patient, consisting of different subpopulations of a-dsDNA, or the same subpopulations as the Wo80, but in different proportions, if tested on the different assays, may give completely different results, ranging from negative to highly positive for the same sample (Fig. 4). The reality that this is happening in real laboratory practice can be nicely demonstrated by having a look at the results of international proficiency testing, such as the UKNEQAS (UK National External Quality Assessment Scheme) [<http://ukneqas.org.uk>].

Since the original Wo/80 standard was utilized by many manufacturers it got depleted. Consequently, a novel international standard became available (Preparation 15/174, www.nibsc.org/products/). Since the new standard is not commutable the original material, a

different reactivity was assigned (100 units).

5. Anti-dsDNA antibodies are relatively specific marker for SLE

SLE is serologically characterized by the presence of antinuclear antibodies (ANA), which is reflected by the inclusion of ANA positivity as one of the immunological criteria for the classification of SLE in both the ACR and the SLICC criteria (Hochberg, 1997; Petri et al., 2012). The number of ANA specificities described in SLE is significantly higher than in other autoimmune diseases, and by 2015 had reached in excess of 180 different autoantibodies (Yaniv et al., 2015). However, the vast majority of these antibodies are probably not specifically linked to SLE, as they occur in other diseases, as well, such as anti-Ro, anti-U1RNP, anti-ssDNA antibodies, and many more. Only very few ANA have proven to be specific for SLE, namely anti-Sm, anti-Rib-P, and to a lesser extent anti-dsDNA antibodies (Kurien and Scofield, 2006; Mahler et al., 2014). However, when testing for a-dsDNA antibodies it needs to be considered that they can also be found in other conditions, even confirmed by several methods (Compagno et al., 2014). A large study using the US military cohort demonstrated that a-dsDNA antibodies can precede the clinical onset of SLE for many years (Arbuckle et al., 2003). Therefore it remains speculative whether these results represent clinically false positive findings or the patients might develop SLE later on. When it comes to a-dsDNA, it can be important to analyze different subpopulations separately. In a patient that is under evaluation for diagnosis at a rheumatologist, a test with high disease specificity is desired. In contrast, in a patient with established SLE under clinical follow-up, an anti-DNA method that is “sensitive-to-change” to identify the risk for renal involvement, or new onset of nephritis and/or active global disease is preferred. Another aspect is the concept of pre-clinical disease which requires a completely different approach, namely highly sensitive screening assays in primary care settings accompanied with adequate confirmation and referral strategies to avoid unnecessary referrals or even misdiagnosis. Although this concept might represent the future of disease prevention, further research is required to fully establish and to validate this concept. It is especially the different avidity of a-dsDNA which determines whether the test result is specific for SLE or not. It has been shown that a-dsDNA as detected by e.g. ELISA, FEIA or ALBIA techniques, measuring both high avidity and low avidity antibodies, are very sensitive for SLE, but exhibit low specificity,

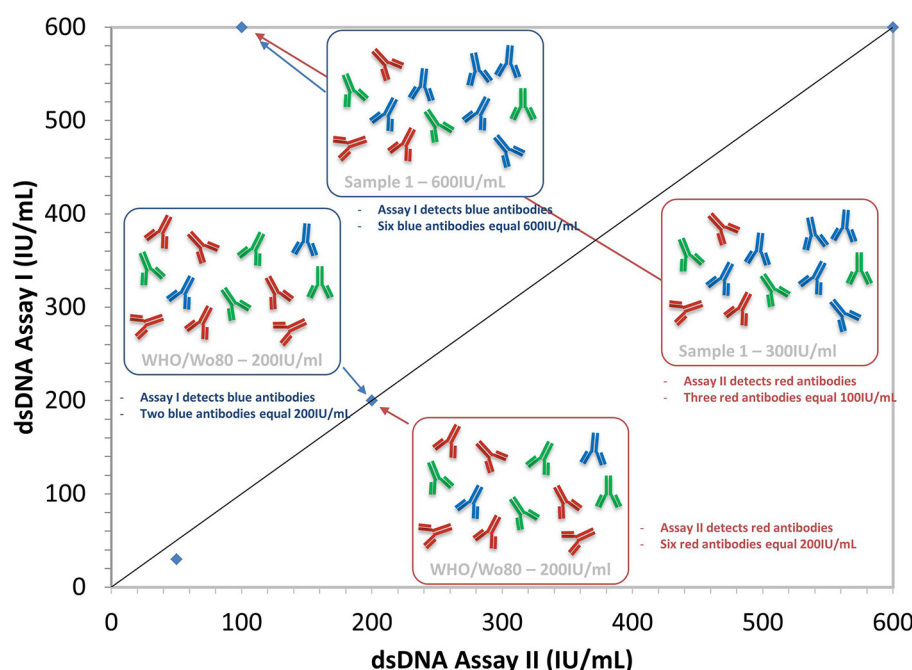


Fig. 4. Lack of correlation of anti-dsDNA antibody assays. The different a-dsDNA populations in the Wo/80 and patient sample are depicted as red, green and blue antibodies. Assay 1 detects the blue antibody population while assay 2 detects the red antibody population. Both assays give 200 IU/ml for the Wo/80 but very different results for the patient sample. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

questioning their diagnostic value (Villalta et al., 2002; Infantino et al., 2015; Smeenk et al., 1990; Smeenk and Hylkema, 1992). However, low avidity a-dsDNA sometimes occur early in the disease course and thus may have some predictive value (Swaak and Smeenk, 1985). Furthermore, in already diagnosed SLE patients, they indicate a milder disease course without renal involvement if present as the only a-dsDNA (Nossent et al., 1989). It has also been described that low affinity a-dsDNA are found more often in SLE patients with central nervous system involvement when compared to high avidity antibodies (Smeenk et al., 1991). The only a-dsDNA which are of high value for the diagnosis of SLE are the high avidity antibodies. In the study of Swaak and Smeenk, assessing 386 non-SLE patients with dsDNA antibodies, 335 (87%) developed SLE, mostly during the first year of entering the study, and the majority of them had high avidity a-dsDNA (Swaak and Smeenk, 1985). The very good correlation of high avidity a-dsDNA with the diagnosis of SLE, together with their predictive value for a more severe disease course, has been confirmed by numerous studies (Werle et al., 1992; Andrejevic et al., 2013).

6. Anti-dsDNA antibody levels correlate with SLE disease activity

One of the first studies showing that a-dsDNA correlate with active SLE was published in 1971 (Koffler et al., 1971), at that time using hemagglutination as the detection method. The authors nicely showed that a-dsDNA titers rise during clinical episodes of SLE activity. These findings were confirmed by numerous publications (Launay et al., 2010; Zen et al., 2012; Narayanan et al., 2010), among them studies that also showed that IgG a-dsDNA, but not IgM a-dsDNA, anti-ssDNA or anti-nucleosome antibodies reliably reflect disease activity and clinical outcomes (Mok et al., 2010; Mok, 2010; Bootsma et al., 1997; Stinton et al., 2007; Mehra and Fritzler, 2014; Li et al., 2015). A more recent study indicated that maybe IgA in addition to IgG can help in the assessment of glomerulonephritis and active disease (Villalta et al., 2013). A significant bias that needs to be considered when assessing the correlation between a-dsDNA and DA is that a-dsDNA are included in many measures for the assessment of DA (e.g. SLEDAI-2K) (Gladman et al., 2002) and clinicians are usually not blinded to the dsDNA result. The good correlation with DA gives rise to some evidence that a-dsDNA are pathogenic (Pisetsky, 2016). This has been shown by several researchers, and will be discussed in the next chapter. The correlation of a-dsDNA levels with DA, however, seems to be strongly dependent on the a-dsDNA detection method used. It is only the high avidity a-dsDNA that correlate well with disease activity, limiting the methods that are useful for this purpose to the Farr RIA, high avidity anti-dsDNA ELISAs, or more recent fully automated tests that are selective for these antibodies, such as CLIA (Bootsma et al., 1997; Andrejevic et al., 2013; Launay et al., 2010; Dervieux et al., 2017). Even if the correlation of high avidity antibodies with DA is highly significant, it should be noted there is a certain patient population in which none of the a-dsDNA tests follows disease activity well enough. These patients may be followed-up with an anti-C1q antibody test as an alternative, which was also shown, similar to a-dsDNA, to associate with DA (Mok et al., 2010; Mok, 2010). In addition, the use of complement consumption (plasma C1q, C3 and C4) together with a-dsDNA levels is a useful approach to better estimate the DA in SLE patients (Sturfelt, 2002; Sturfelt and Truedsson, 2005). However, it should be emphasized that also C3, C4 and CH50 is included as an item in SLEDAI-2K (Gladman et al., 2002). In line with their correlation with DA, a-dsDNA have been shown to be able to predict relapses (Bootsma et al., 1997). The question that is not answered yet is if their predictive value is sufficient to base a treatment decision on the a-dsDNA level (Launay et al., 2010; Andrejevic et al., 2013). From a different point of view, in times of low DA, sustained reductions in a-dsDNA regardless of the treatment, was shown to lead to clinically meaningful improvements in patient-reported health-related quality of life (Strand and Crawford, 2005).

7. Anti-dsDNA antibodies constitute a risk factor for renal involvement

Approximately 30–60% of SLE patients suffer from LN in the course of the disease, with a higher prevalence in Asians, Hispanics, Native Americans, and African descendants, especially in females of child-bearing age (Yung and Chan, 2015). Besides the geographic differences, the cohort composition also impacts the prevalence of LN in SLE. Patients from tertiary university centers, often show a higher prevalence of patients with LN compare to primary or secondary care centers. Since LN is a major determinant of disease morbidity and mortality in SLE patients (Cervera et al., 2003), a marker that is useful for risk assessment is needed. Identifying LN patients early and utilizing the ‘window of opportunity’ prevents progression to end-stage renal disease (Hanly et al., 2016). Anti-dsDNA antibodies can serve as such a marker, as they were reported to be an independent risk factor for renal involvement. Moroni and colleagues showed that high titers of a-dsDNA or anti-C1q antibodies were independent predictors that discriminate proliferative from non-proliferative LN at the time of renal biopsy ($p = 0.005$, OR = 8.67, CI: 2.03–37.3) (Moroni et al., 2015). The authors concluded that anti-C1q antibodies alone or in combination with a-dsDNA emerged as the most reliable test in differentiating the two types of LN. The combination of anti-C1q antibodies with a-dsDNA was confirmed the highest risk factor for renal survival, higher than the single antibodies that did not reach statistical significance (Yang et al., 2012). Again, just like for the diagnosis of SLE and for assessing DA, it is the high avidity antibodies that are closely linked to LN (Andrejevic et al., 2013). The correlation of a-dsDNA with the risk for LN has been confirmed by many different groups (Isenberg, 1997; Narayanan et al., 2010; Isenberg et al., 1997; Linnik et al., 2005; Ravirajan et al., 2001), and is closely linked to the pathogenicity of a-dsDNA in general, and to their role in the pathogenesis of LN in particular. In 1967 Koffler and colleagues provided evidence that a-dsDNA could be eluted from the kidneys of lupus patients (Koffler et al., 1967), which was a hint implicating but not proving full evidence for the pathogenicity of these antibodies. The missing compelling evidence was provided by animal model experiments of Raz and co-workers, who used an isolated rat kidney perfusion system and showed clearly that murine a-dsDNA were capable of increasing the amount of proteinuria produced by the kidneys significantly (Raz et al., 1989). There are currently different hypotheses on how a-dsDNA may act as pathogen, and all of them are based on the binding of these antibodies or immune-complexes containing these antibodies to kidney constituents, as convincingly shown in mouse models (van Bavel et al., 2007), and subsequent internalization of a-dsDNA into mesangial cells. Accumulating evidence suggests a-dsDNA contribute to the pathogenesis of LN either through binding of immune complexes and thus triggering complement activation, or through direct or indirect binding to cross-reactive antigens or chromatin materials, respectively, to resident renal cells and/or extracellular matrix components (Yung and Chan, 2015; Yung et al., 2015), thereby triggering downstream cellular activation and proliferation as well as inflammatory and fibrotic processes.

8. Anti-dsDNA antibodies in ANA negative SLE

Antinuclear antibodies (ANA) measured by indirect immunofluorescence (IIF) on HEP-2 cell substrates are a very sensitive marker for SLE. The vast majority of SLE patients are reported to show a positive ANA test at least some time during their disease course, but the point-prevalence may be significantly lower (Tan et al., 1982; Fessel, 1978; Sjowall et al., 2008; Wirestam et al., 2015; Acosta-Merida and Isenberg, 2013). The exact fraction of SLE patients positive for ANA depends on several factors including but not limited to the assay/cell substrate used, the screening dilution, the cohort composition (severe or mild SLE) as well as the definition of ANA (cytoplasmic staining included or excluded). However, there is a certain proportion of SLE

patients, ranging from 1% to 20% (Tan et al., 1982), that are ANA negative in IIF on HEp-2 cell substrates (Raz et al., 1988). Even though seronegative SLE exists, most of these samples are not negative when tested on solid phase assays, a proportion showing positivity for anti-SS-A/Ro60 antibodies, due to the well described low sensitivity of IIF for these antibodies (Reichlin, 2000; Provost and Reichlin, 1981; Perez et al., 2017), while some others show positivity for anti-Rib-P (Mahler et al., 2008; Mahler et al., 2014), or dsDNA antibodies (Zhao, 2016; Lindstedt et al., 1977; Gladman et al., 1978). Thus, detection of anti-dsDNA antibodies by a single analyte tests, even in ANA negative cases, is very important. In addition to the identification of more SLE patients, the test provides help in risk assessment for clinical complications. It has been shown that the subgroup of SLE patients that are negative for ANA in IIF, but positive for dsDNA, tend to have more severe complications, among them nephritis (Lindstedt et al., 1977), dystrophic calcification (Morris et al., 1998), or severe autoimmune neutropenia (Zhao, 2016).

9. One test for anti-dsDNA antibodies may not be sufficient in clinical practice

Available a-dsDNA tests range from insensitive but very specific (CLIFT) to very sensitive but unspecific (normal ELISA). Before choosing a specific a-dsDNA test, the laboratory scientist should consider the testing environment (primary care or secondary/tertiary care), and answer the question, ‘what is the intended use of the assay?’. In a secondary/tertiary care environment, where primarily specialist physicians order the test (in some jurisdictions general practitioners do not order ANA whereas in others they constitute the vast majority of ANA test requesters), and the pre-test probability is high, a high sensitivity assay may be preferred in order to support diagnosis and initiate treatment. In a setting of companion medicine, certain a-dsDNA tests may be used as a selection criterion for certain SLE drug trials (Pisetsky et al., 2017). The resulting lower specificity of such an assay can be dealt with by the clinicians, because they make a diagnosis primarily upon clinical symptoms, and the a-dsDNA test is rather used for confirmation than for differential diagnosis. In this setting, a combination of different a-dsDNA assays may provide more meaningful clinical information and allow a more comprehensive interpretation in the clinical context. However, in a primary care environment with a very low pre-test probability where the clinical paradigm is case finding rather than intent to treat (Fritzler, 2016), a low specificity will have a dramatic effect on the number of false positive results generated. In fact, the vast majority of positive results will be false positive (Mahler et al., 2014). This would be in favor of a highly specific test, such as CLIFT. However, since CLIFT has a low sensitivity, SLE patients may be missed, resulting in delayed diagnosis and treatment, and clinical deterioration. In addition, it needs to be emphasized here that many other autoantibodies are tested in the diagnosis of SLE and autoantibodies are just an aid in the diagnosis. Consequently, missing a diagnosis of SLE due to a negative CLIFT result is rare. Thus, the combination of a sensitive and a highly specific assay may be recommended. The sensitive assay should not be too sensitive (and therefore unspecific), though, as the interpretation of a positive result in the sensitive assay and a negative result in the specific assay will be very difficult. A possible solution is to choose the combination of a sensitive test that has a high specificity, as well, such as a CLIA assay. In this case a positive result in the sensitive and specific assay, and a negative result in the highly specific assay can still be interpreted as high suspicion of SLE, and should give rise to referral of the patient to a rheumatologist.

The results of a-dsDNA tests are used to support diagnosis of suspected SLE, but they are also used in monitoring SLE disease activity. If the a-dsDNA test is requested upon a suspicion of SLE, a high sensitivity may be required in order not to miss any patients. However, since clinical symptoms are often vague, the combination of a sensitive and a specific test may be recommended in this situation, as outlined above

(Isenberg et al., 1987). When it comes to monitoring patients under treatment, a test should be chosen that parallels disease activity. This can be done by different tests with different precision, but excludes the CLIFT assay, because titers may stay high even though disease activity decreases substantially (Andrejevic et al., 2013; Zhao et al., 2018). Once SLE is diagnosed, the clinician may also want to assess the risk for nephritis. Only a couple of tests have proven to correlate with SLE nephritis (Bentow et al., 2016; Linnik et al., 2005) and limited data are available to assess the predictive value for future development of LN, which is one reason for the potential requirement of a second test in addition to the standard a-dsDNA test used for screening or to support diagnosis.

10. The guidelines do not recommend specific methods

The existing guidelines for classification of SLE include a-dsDNA as one criterion, with certain differences. The 1982 ACR revised criteria for classification of SLE include “anti-DNA antibody to native DNA in abnormal titer” and anti-Sm antibodies as immunological criteria. The 1997 revised criteria also include anti-phospholipid antibodies (Tan et al., 1982). The more recent SLICC (systemic lupus international collaborating clinics) classification criteria for SLE require “anti-dsDNA above laboratory reference range, except ELISA: twice above laboratory reference range” (Petri et al., 2012), and thus appreciates the reported low specificity of ELISA a-dsDNA tests, particularly of the low positive results. International recommendations for the assessment of autoantibodies to cellular antigens referred to as anti-nuclear antibodies, published in 2014, state that “the Farr assay and the CLIFT offer high clinical specificity (Agmon-Levin et al., 2014). Alternative methods may yield lower specificity and, if so, it was recommended that positive results obtained by these methods be confirmed by CLIFT or Farr assay—and be reported separately” (Agmon-Levin et al., 2014). In summary, these guidelines do not recommend a certain method for a-dsDNA detection, but two of them at least take the different assay characteristics into account, and the most recent one promotes the use of two different methods to be on the safe side.

Competing interest

E. Mummert, M. Mahler, and C. Bentow are employees of Inova Diagnostics. M. Fritzler has received consultant fees from Inova Diagnostics, BioRad and gifts in kind from Euroimmun GmbH.

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